

Problem Set for 07 April 2025

Example (Quotient Spaces in Analysis).

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be of period T . Then there exists a unique function $\tilde{f} : (\mathbb{R}/T) \rightarrow \mathbb{R}$ mapping the coset $[x] := x + T \cdot \mathbb{Z}$ to $f(x)$. Explicitly,

$$\tilde{f} : (\mathbb{R}/T) \rightarrow \mathbb{R}, \quad (x + T \cdot \mathbb{Z}) \mapsto f(x).$$

2. The derivation $\frac{d}{dx} : D(\mathbb{R}) \rightarrow \mathbb{R}^{\mathbb{R}}$ maps differentiable functions f to f' . There exists a unique linear map

$$D(\mathbb{R})/\mathbb{R} \rightarrow \mathbb{R}^{\mathbb{R}}, \quad (f + \mathbb{R} \cdot \text{id}) \mapsto f'.$$

3. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function satisfying $f(x, y) = f(x - t, y - t)$ for all $x, y, t \in \mathbb{R}$. Then f is automatically defined on $\mathbb{R}^2/\text{span}((1, 1)) \simeq \mathbb{R}$.

4. etc.

Exercise Let V be a vector space over $\mathbb{F}$, and define $\text{End}(V) := \text{Hom}_{\mathbb{F}}(V, V)$. Define:

- $f \in \text{End}(V)$ as *almost zero* (denoted $f \in \text{AZ}(V)$), provided $\dim \text{im}(f) < \infty$.
- $f \in \text{End}(V)$ as *almost an isomorphism* (denoted $f \in \text{AI}(V)$), provided $\dim \ker f + \dim \frac{V}{\text{im}(f)} < \infty$.

Now finish the following.

1. If $\dim V < \infty$, determine $\text{AZ}(V)$ and $\text{AI}(V)$.
2. If $V = \mathbb{R}[x]$, find an injective $f \in \text{AI}(V)$ and a surjective $g \in \text{AI}(V)$. Additionally, find some $h \notin \text{AI}(V) \cup \text{AZ}(V)$.
3. Determine whether $\text{AZ}(V)$ is a subspace of $\text{End}(V)$. What about $\text{AI}(V)$?
4. Show that for arbitrary $f, g \in \text{AZ}(V)$, one has $g \circ f \in \text{AZ}(V)$.
5. Prove that $\text{AZ}(V)$ is an **ideal** in $\text{End}(V)$. The term **ideal** is defined in 20 March 2025.
6. Show that for arbitrary $f, g \in \text{AI}(V)$, one has $g \circ f \in \text{AI}(V)$.
7. Demonstrate that if any two elements of $\{f, g, g \circ f\}$ belong to $\text{AI}(V)$, then so does the third.

Hint: For $\Phi(f) := \dim \ker f + \dim \frac{V}{\text{im}(f)} < \infty$, show that $\Phi(f \circ g) = \Phi(f) + \Phi(g)$.

8. Show that for arbitrary $f \in \text{AZ}(V)$, one has $(\text{id}_V + f) \in \text{AI}(V)$.
9. **(Optional)** By Axiom of Choice, every subspace has a complementary subspace. Prove that $f \in \text{AI}(V)$ can always be expressed as the summation of a map of $\text{AZ}(V)$ -type and an automorphism (i.e., auto = iso + endo). Conversely, show that any such summation belongs to $\text{AI}(V)$.
10. **(Optional)** Show that $\frac{\text{End}(V)}{\text{AZ}(V)}$ is a quotient ring with a well-defined subset $\frac{\text{AI}(V)}{\text{AZ}(V)}$, consisting precisely of the invertible elements. In this quotient ring, the operators $\frac{d}{dx}(-)$ and $\int_0^x (-) dx$ are mutually inverse.

Problem (Optional) Let V be a vector space over $\mathbb{F}$. We define

$$\prod_{i \in I} V := V \times \underbrace{\cdots \times}_{I\text{-copies}} V \simeq \text{Hom}_{\text{Sets}}(I, V)$$

as an $\mathbb{F}$ -vector space whose elements are sequences $(v_i)_{i \in I}$. Informally, $\prod_{i \in I} V$ represents the space of sequences indexed by I with values in V .

1. Define $e_i := (\dots, 0, 0, \underbrace{1_{\mathbb{F}}}_{i\text{-th entry}}, 0, 0, \dots) \in \prod_{i \in I} \mathbb{F}$. Determine when $\{e_i\}_{i \in I}$ forms a basis of $\prod_{i \in I} \mathbb{F}$.
2. Describe $\text{span}(\{e_i\}_{i \in I})$ in natural language.
3. Substituting I with $V_{\text{as a set}}$, the underlying set of V as an $\mathbb{F}$ -vector space, prove that π is a linear surjection and determine the equivalence relation $R_{\mathbb{F}}$ as defined in 24 March 2025:

$$\pi : \text{span}(\{e_v\}_{v \in V_{\text{as a set}}}) \rightarrow V, \quad e_v \mapsto v.$$